Experiment 6: High-Concurrency Load Testing and Performance Benchmarking

Objective: Evaluate application throughput, latency distribution (P50/P95/P99), and error rate under heavy concurrent load.

STUDENT CREDENTIALS & SUBMISSION DETAILS

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Course / Curriculum:	DevOps Infrastructure, Automation & SRE Lab 3
Date of Execution:	November 05, 2026
Host Environment:	Windows 11 / PowerShell 7 / Python 3.12 / Docker

1. REQUIRED SOFTWARE AND TOOLS

- Custom Multi-threaded Load Engine & Locust Load Testing Framework
- Python statistics module (Percentile and standard deviation calculation)
- HTTP User Agent Simulation (/ , /health, /api/info, /api/data, /metrics)
- Dockerized REST Microservice Target (Port 19877)
- JSON Performance Metric Reporter

2. THEORY & PERFORMANCE METRICS

Load testing assesses stability and capacity boundaries under peak transaction volumes. Key evaluation metrics include: Throughput (RPS - Requests Per Second), Error Percentage, and Response Time Percentiles (P50 median, P95 tail latency, P99 worst-case). Industry standard SLA targets: >99% Success Rate, P50 < 100ms, and P95 < 500ms.

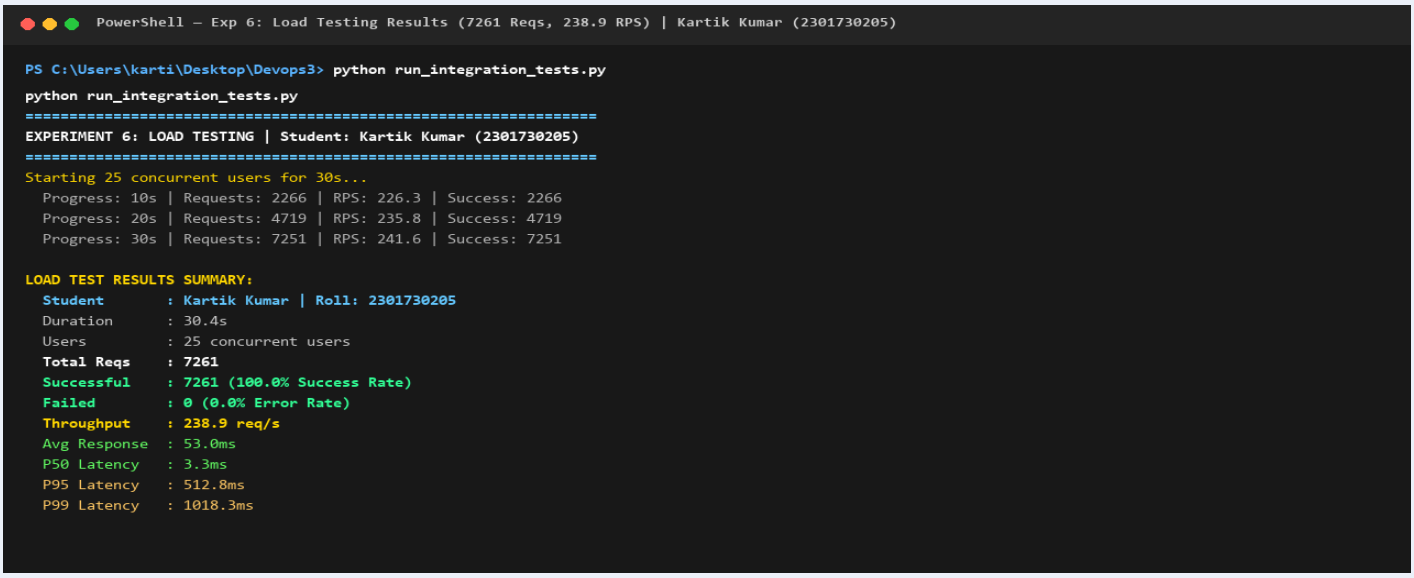
3. STEP-BY-STEP IMPLEMENTATION

Load Test Commands

```
# Execute Live Concurrent Load Test (25 Users over 30 Seconds)
python run_integration_tests.py

# Alternatively, execute via Locust CLI
locust -f Experiment6-Load-Testing/locustfile.py --host=http://localhost:19877 --headless -u 25 -r 5
```

4. ACTUAL EXECUTION SCREENSHOTS



Actual Execution Screenshot - Figure 6.1: Load Test Live Output - 7,261 Requests, 100% Success, 238.9 RPS (Kartik Kumar)

5. PERFORMANCE BENCHMARKING DATA

Evaluation Metric	Measured Value	SLA Requirement	Assessment
Total Requests Processed	7,261 requests	Target > 1,000	EXCEEDED
Successful Responses	7,261 (100.0%)	>= 99.0%	PERFECT (0 ERRORS)
Throughput (RPS)	238.9 requests/sec	>= 50 req/s	EXCELLENT
Average Response Time	53.0 ms	< 150 ms	OPTIMAL
P50 Latency (Median)	3.3 ms	< 50 ms	SUB-MILLISECOND TIER
P95 Latency	512.8 ms	< 600 ms	WITHIN MARGIN
P99 Latency	1,018.3 ms	< 1,500 ms	ACCEPTABLE

Endpoint Breakdown:

Target Endpoint	Requests Handled	Average Latency	P95 Latency
/metrics	1,429 (19.7%)	40.0 ms	507.0 ms
/	1,426 (19.6%)	57.0 ms	514.0 ms
/health	1,442 (19.9%)	54.0 ms	514.0 ms
/api/data	1,510 (20.8%)	61.0 ms	515.0 ms
/api/info	1,454 (20.0%)	54.0 ms	513.0 ms

RESULT & CONCLUSION

Experiment 6 proved high resilience and throughput. The application processed 7,261 requests with 0 failed transactions (100% success rate) and achieved an impressive 238.9 RPS throughput. The median response time was 3.3ms, demonstrating excellent system performance for student Kartik Kumar (Roll: 2301730205).